

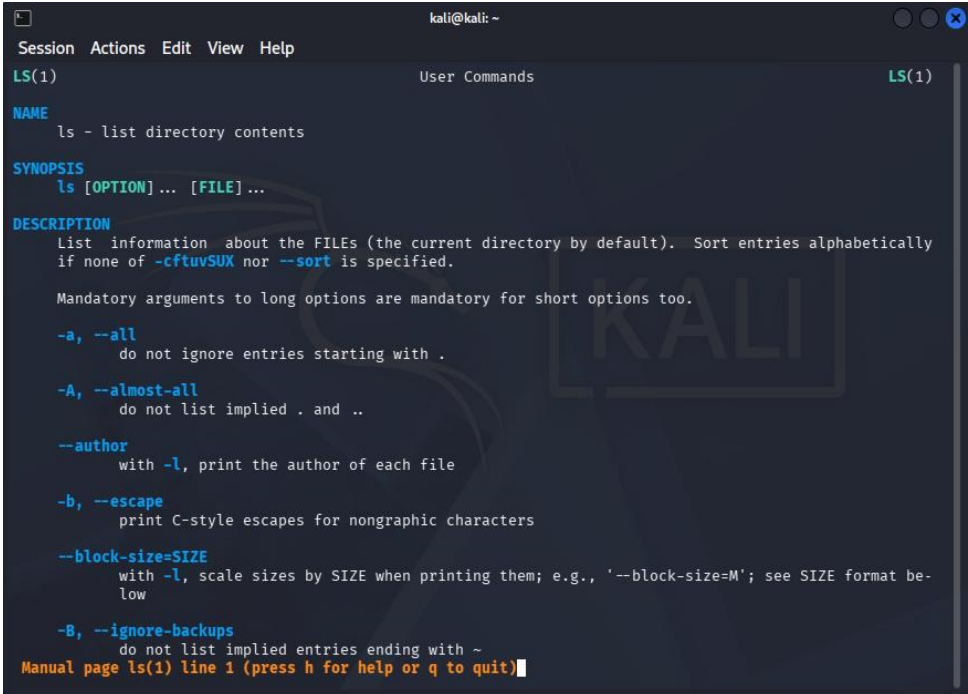
LOS_Lab Assignment 2

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The main aim of this lab session is to provide hands-on experience on:

1. Getting Help

Task	Command Name	Syntax	Example
To get manual page for the known command	man	man <command>	man ls
			
To get manual page for the unknown command	apropos	apropos	apropos copy

```

kali@kali: ~
Session Actions Edit View Help
pgmtopgm (1) - copy PGM image
pnmtopnm (1) - copy a PNM image
pmtoppm (1) - copy PPM image
ptrepack (1) - Copy any PyTables Leaf, Group or complete subtree into another file.
rsync (1) - a fast, versatile, remote (and local) file-copying tool
scp (1) - OpenSSH secure file copy
scp1 (1) - secure copy (remote file copy program)
ssh-copy-id (1) - use locally available keys to authorise logins on a remote machine
stpcpy (3) - string return-offset-pointer copy
strcat (3) - copy or concatenate a string
strcpy (3) - copy or concatenate a string
string_copying (7) - copying strings and character sequences
svnversion (1) - Produce a compact version identifier for a working copy.
UFFDIO_COPY (2const) - atomically copy a continuous memory chunk into the userfault registered range
uuid_copy (3) - copy a UUID value
va_copy (3) - variable argument lists
vfs_shadow_copy (8) - Expose snapshots to Windows clients as shadow copies.
vfs_shadow_copy2 (8) - Expose snapshots to Windows clients as shadow copies.
wcpncpy (3) - copy a wide-character string, returning a pointer to its end
wcpncpy (3) - copy a fixed-size string of wide characters, returning a pointer to its end
wcscpy (3) - copy a wide-character string
wcscpy (3) - copy a fixed-size string of wide characters
wmemncpy (3) - copy an array of wide-characters
wmemmove (3) - copy an array of wide-characters
wmemncpy (3) - copy memory area
x86_64-linux-gnu-objcopy (1) - copy and translate object files
x86_64-w64-mingw32-objcopy (1) - copy and translate object files
xclip-copyfile (1) - copy and move files via the X clipboard
xclip-cutfile (1) - copy and move files via the X clipboard
xclip-pastefile (1) - copy and move files via the X clipboard

(kali@kali)-[~]
$

```

To know the source file binary	whereis	whereis <command>	whereis ls
<pre> (kali@kali)-[~] \$ whereis ls ls: /usr/bin/ls /usr/share/man/man1/ls.1.gz </pre>			
To know the path of the command	which	which <command>	which cd
<pre> (kali@kali)-[~] \$ which cd cd: shell built-in command </pre>			
To know the command is external or internal	type	type<command>	type cd
<pre> (kali@kali)-[~] \$ type cd cd is a shell builtin </pre>			
To get help for the internal command	help	help <command>	help cd

```

$ help cd
cd: cd [-L|[-P [-e]]] [-@] [dir]
    Change the shell working directory.

    Change the current directory to DIR. The default DIR is the value of the
    HOME shell variable. If DIR is "-", it is converted to $OLDPWD.

    The variable CDPATH defines the search path for the directory containing
    DIR. Alternative directory names in CDPATH are separated by a colon (:).
    A null directory name is the same as the current directory. If DIR begins
    with a slash (/), then CDPATH is not used.

    If the directory is not found, and the shell option `cdable_vars' is set,
    the word is assumed to be a variable name. If that variable has a value,
    its value is used for DIR.

    Options:
    -L      force symbolic links to be followed: resolve symbolic
            links in DIR after processing instances of `..'
    -P      use the physical directory structure without following
            symbolic links: resolve symbolic links in DIR before
            processing instances of `..'
    -e      if the -P option is supplied, and the current working
            directory cannot be determined successfully, exit with
            a non-zero status
    -@      on systems that support it, present a file with extended
            attributes as a directory containing the file attributes

    The default is to follow symbolic links, as if `-L' were specified.
    `..' is processed by removing the immediately previous pathname component
    back to a slash or the beginning of DIR.

    Exit Status:
    Returns 0 if the directory is changed, and if $PWD is set successfully when
    -P is used; non-zero otherwise.

```

To list out bash
commands

compgen

compgen - c

compgen -c

```

$ compgen -c
diff
egrep
fgrep
grep
ip
l
la
ll
ls
if
then
else
elif
fi
case
esac
for
select
while
until
do
done
in
function
time
{
}
!
[[
]]
coproc

```

To know the usage of
the command

whatis

whatis <command>

whatis ls

```

(kali@kali)-[~]
$ whatis ls
ls (1)          - list directory contents

```

2. Basic commands

Task	Command Name	Syntax	Example
To know today's date	date	date	date
<pre> \$ date Thursday 20 August 2026 04:22:05 PM UTC </pre>			
To print calendar	cal	cal [month] [year]	cal
<pre> \$ cal 8 2026 August 2026 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 </pre>			
To print kernel version	uname	uname -r	uname -r
<pre> (kali@kali)-[~] \$ uname -r 6.19.14+kali-amd64 </pre>			
To print default shell	echo	echo \$SHELL	echo \$SHELL
<pre> (kali@kali)-[~] \$ echo \$SHELL /usr/bin/zsh </pre>			
To print currently logged in user	whoami	whoami	whoami
<pre> (kali@kali)-[~] \$ whoami kali </pre>			

To create shortcut for command	alias	alias name='command'	alias l='ls -l'
<pre>(kali@kali)~\$ alias ll='ls -l'</pre>			
To delete shortcut	unalias	unalias name	unalias l
<pre>(kali@kali)~\$ unalias l</pre>			
To change the timestamp of the file	touch	touch <file>	touch -m time2.txt
<pre>(kali@kali)~\$ touch -m time2.txt</pre>			
To create empty files	touch	touch file1 file2	touch lab2.txt lab2.c
<pre>(kali@kali)~\$ touch lab2.txt lab2.c</pre>			
To clear the screen	clear	clear	clear
<pre>(kali@kali)~\$ clear</pre>			
To know disk usage	du	du [options] [path]	du -sh .
<pre>(kali@kali)~\$ du -sh . 4.0K .</pre>			
To know free space in the system	df	df [options]	df -h

```

$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            871M   0    871M   0% /dev
tmpfs           194M  1.2M  193M   1% /run
/dev/sda1       79G   16G   59G  22% /
tmpfs           967M  4.0K  967M   1% /dev/shm
none            1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           967M  8.0K  967M   1% /tmp
none            1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           194M  108K  194M   1% /run/user/1000

```

To know about the linux release

cat /etc/os-release

cat /etc/os-release

cat /etc/os-release

```

$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.2"
VERSION="2026.2"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"

```

To show login history

last

last

last

```

$ last
kali      tty7      :0                Tue Aug 25 05:39 - 01:26 (213503982+03:48)
lightdm   tty7      :0                Tue Aug 25 05:39 - 05:39 (00:00)
kali      tty7      :0                Tue Aug 25 01:31 - still logged in
lightdm   tty7      :0                Tue Aug 25 01:31 - 01:31 (00:00)
kali      tty7      :0                Tue Aug 18 01:10 - still logged in
lightdm   tty7      :0                Tue Aug 18 01:10 - 01:10 (00:00)
kali      tty7      :0                Tue Aug 18 00:34 - 01:09 (00:34)
lightdm   tty7      :0                Tue Aug 18 00:34 - 00:34 (00:00)
lightdm   tty7      :0                Fri Aug 14 04:52 - still logged in
kali      tty7      :0                Wed Aug 12 17:48 - still logged in
lightdm   tty7      :0                Wed Aug 12 17:48 - 17:48 (00:00)
kali      tty7      :0                Tue Aug 11 02:54 - still logged in
lightdm   tty7      :0                Tue Aug 11 02:54 - 02:54 (00:00)
kali      tty7      :0                Mon Aug 10 19:31 - 01:49 (06:18)
lightdm   tty7      :0                Mon Aug 10 19:31 - 19:31 (00:00)
postgres          Tue Jun 16 10:29 - 10:29 (00:00)

wtmpdb begins Tue Jun 16 10:29:31 2026

```

To check the current user privileges

sudo

sudo -l

sudo -l

```

└─$ sudo -l
[sudo] password for kali:
Matching Defaults entries for kali on kali:
    secure_path=/usr/sbin\:/usr/bin\:/sbin\:/bin

Runas and Command-specific defaults for kali:
    Defaults!/usr/sbin/visudo env_keep+="SUDO_EDITOR EDITOR VISUAL"

User kali may run the following commands on kali:
    (ALL : ALL) ALL

```

To find about the Linux release

lsb_release

lsb_release -a

lsb_release -a

```

└─$ lsb_release -a
No LSB modules are available.
Distributor ID: Kali
Description:    Kali GNU/Linux Rolling
Release:        2026.2
Codename:       kali-rolling

```

To check system uptime

uptime

uptime

uptime

```

└─$ uptime
02:06:07 up 31 min, 1 user, load average: 0.01, 0.08, 0.08

```

To shutdown

shutdown

sudo shutdown -h now

sudo shutdown -h now

```

└─$ sudo shutdown -h now

```

To restart

reboot

sudo reboot

sudo reboot

```

(kali@kali)-[~]
└─$ sudo reboot

```

3. Navigation

Task	Command	Syntax	Example
To navigate home directory	cd	cd ~	cd ~
To navigate to the parent directory	cd	cd ..	cd ..
To navigate to the child directory	cd	cd <directory>	cd Lab2
Alternate command to cd	pushd	pushd <directory>	pushd Lab2
To go back to the previous directory	cd -	cd -	cd -
To go to the root directory	cd	cd /	cd /

```
(kali㉿kali)-[~]
$ cd ~

(kali㉿kali)-[~]
$ cd ..

(kali㉿kali)-[/home]
$ cd -
~

(kali㉿kali)-[~]
$ cd /

(kali㉿kali)-[/]
$
```

4. File System

Task	Syntax	Command
How to identify the file system	df -T [path]	df -T .

```
(kali㉿kali)-[/]  
$ df -T  
Filesystem      Type      1K-blocks    Used Available Use% Mounted on  
udev            devtmpfs   891096         0    891096   0% /dev  
tmpfs           tmpfs     197964        1184    196780   1% /run  
/dev/sda1       ext4     82083148 16531328 61336272 22% /  
tmpfs           tmpfs     989820         4    989816   1% /dev/shm  
none            tmpfs      1024          0      1024   0% /run/credentials/systemd-journald.service  
tmpfs           tmpfs     989820         8    989812   1% /tmp  
none            tmpfs      1024          0      1024   0% /run/credentials/getty@tty1.service  
tmpfs           tmpfs     197964        108    197856   1% /run/user/1000
```

5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

No	Task	Pattern
a	List all files ending in .html.	ls *.html
b	List all files other than .html files using extended globbing.	ls !(*.html)
c	List filenames containing at least one numeric digit.	ls *[0-9]*
d	List files whose names begin with user followed by a single digit.	ls user[0-9].*
e	List files ending in either .txt or .log using a single extended glob pattern.	ls +(*.txt *.log)
f	List files that do not end in .txt or .log.	ls *.{txt,log}
g	List files beginning with either access or error using extended globbing.	ls {access*,error*}
h	List .html files whose filenames contain a numeric digit.	ls *[0-9]*.html

Result:

```
(kali㉿kali)-[~]
└─$ ls *.html
admin.html  backup1.html  index.html  login.html

(kali㉿kali)-[~]
└─$ ls !(*.html)
zsh: unknown file attribute: h

(kali㉿kali)-[~]
└─$ ls *[0-9]*
backup1.html  backup2.txt  file123  lab2.c  lab2.txt  time2.txt  user1.txt  user2.log  user2.txt

(kali㉿kali)-[~]
└─$ ls *.{txt,log}
access.log  backup2.txt  error.log  lab2.txt  report.txt  time2.txt  user1.txt  user2.log  user2.txt

(kali㉿kali)-[~]
└─$ ls {access*,error*}
access.log  error.log

(kali㉿kali)-[~]
└─$ ls *[0-9]*.html
backup1.html
```

6. Extended Globbing

1. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

No	Task	Pattern
a	Write an extended glob pattern that selects only access.log, error.log, and auth.log.	ls @(access.log error.log auth.log)
b	Select all files except notes.txt.	ls !(@(notes.txt))
c	Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.	ls auth.log.@(1 2)

2. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log

No	Task	Pattern
a	construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.	ls malware*[0-9]*.log

3. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log

No	Task	Pattern
a	use extended globbing to select .log files containing either failed_login or successful_login.	ls @(failed_login successful_login).log
b	Construct a pattern that lists all .log files except firewall.log.	ls !(@(firewall.log))

7.Filters

No	Task	Command
1	Display the first 10 entries of login.log.	head -n 10 login.log
2	Display the first 5 entries of access.log	head -n 5 access.log
3	Display the first 15 security events from security.log.	head -n 15 security.log
4	Display the first 20 events recorded in incident.log.	head -n 20 incident.log
5	Display only the first entry in login.log.	head -n 1 login.log

6	Display the first 25 web requests recorded in access.log.	head -n 25 access.log
7	Display the last 10 entries of login.log.	tail -n 10 login.log
8	Display the last 5 entries of access.log.	tail -n 5 access.log
9	Display the 20 most recent security events from security.log.	tail -n 20 security.log
10	Display the 25 most recent events from incident.log.	tail -n 25 incident.log
11	Display only the last entry in security.log.	tail -n 1 security.log
12	Display the last 50 events recorded in incident.log.	tail -n 50 incident.log
13	Display the contents of login.log starting from line 41 until the end of the file.	tail -n +41 login.log
14	Display the contents of security.log starting from line 101 until the end of the file.	tail -n +101 security.log
15	Find the total number of lines in login.log.	wc -l login.log
16	Find the total number of lines in access.log.	wc -l access.log
17	Find the total number of security events in security.log.	wc -l security.log
18	Determine the total number of events recorded in incident.log.	wc -l incident.log

19	Count the number of words in login.log.	wc -w login.log
20	Count the number of bytes in security.log.	wc -c security.log
21	Display the line, word, and byte counts of access.log.	wc -l -w -c access.log
22	Find the line counts of all four .log files using a single wc command.	wc -l *.log
23	Display the first 20 entries of login.log and, from those entries, display only the last 5.	head -n 20 login.log tail -n 5
24	Display the last 20 entries of login.log and, from those entries, display only the first 5.	tail -n 20 login.log head -n 5
25	Display the first 30 entries of access.log and then display only the last 10 entries from that output	head -n 30 access.log tail -n 10
26	Display the last 30 entries of access.log and then display only the first 10 entries.	tail -n 30 access.log head -n 10
27	Display the first 50 events from security.log and then display the last 10 events from those 50.	head -n 50 security.log tail -n 10
28	Display the first 50 events from security.log and then display the first 10 events from those 50.	tail -n 50 security.log head -n 10
29	Display entries from line 11 to line 20 of login.log.	tail -n +11 login.log head -n 10 or sed -n '11,20p' login.log
30	Display entries from line 21 to line 30 of login.log.	tail -n +21 login.log head -n 10 or sed -n '21,30p' login.log

31	Display entries from line 11 to line 20 of access.log.	tail -n +11 access.log head -n 10 or sed -n '11,20p' access.log
32	Display entries from line 31 to line 40 of access.log.	tail -n +31 access.log head -n 10 or sed -n '31,40p' access.log
33	Display security events from line 21 to line 30 of security.log.	tail -n +21 security.log head -n 10 or sed -n '21,30p' security.log
34	Display security events from line 51 to line 60 of security.log.	tail -n +51 security.log head -n 10 or sed -n '51,60p' security.log
35	Display incident events from line 101 to line 110 of incident.log.	tail -n +101 incident.log head -n 10 or sed -n '101,110p' incident.log
36	Display incident events from line 201 to line 220 of incident.log.	tail -n +201 incident.log head -n 20 or sed -n '201,220p' incident.log
37	Display incident events from line 491 to line 500.	tail -n +491 incident.log head -n 10 or sed -n '491,500p' incident.log
38	Take the first 20 entries of login.log and count how many lines are displayed.	head -n 20 login.log wc -l
39	Take the last 15 entries of access.log and count how many lines are displayed.	tail -n 15 access.log wc -l

40	Take the first 50 entries of security.log and count the number of words in those entries.	head -n 50 security.log wc -w
41	Take the last 25 entries of security.log and count the number of lines.	tail -n 25 security.log wc -l
42	Extract lines 21–30 from login.log and count the number of extracted lines.	tail -n +21 login.log head -n 10 wc -l or sed -n '21,30p' login.log wc -l
43	Extract lines 51–60 from access.log and count the number of extracted lines.	tail -n +51 access.log head -n 10 wc -l or sed -n '51,60p' access.log wc -l
44	Extract events 101–150 from incident.log and count how many events are displayed	tail -n +101 incident.log head -n 50 wc -l or sed -n '101,150p' incident.log wc -l
45	Display the first 100 events from incident.log, take the last 20 events from that output, and count the number of lines.	head -n 100 incident.log tail -n 20 wc -l

